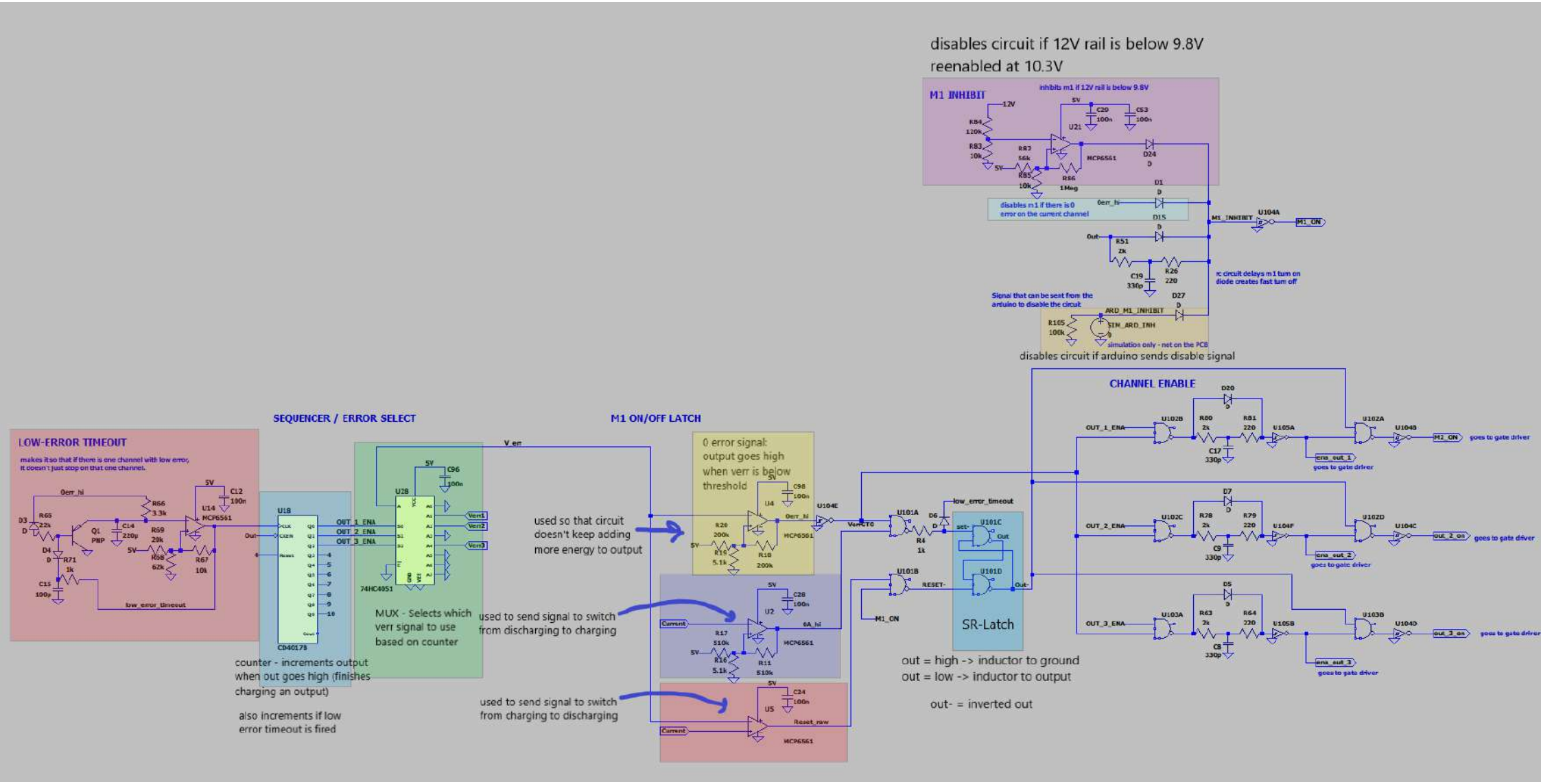
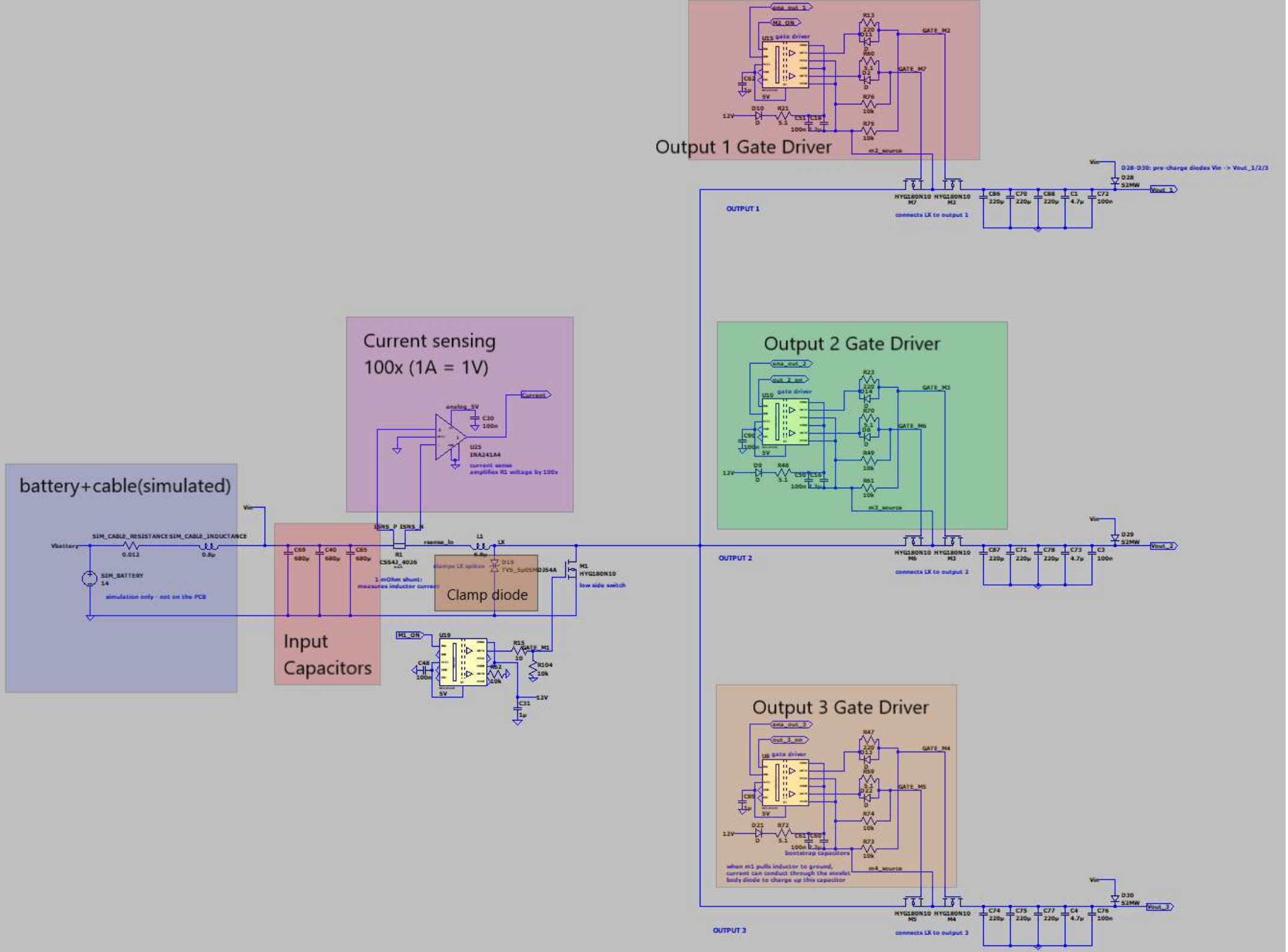
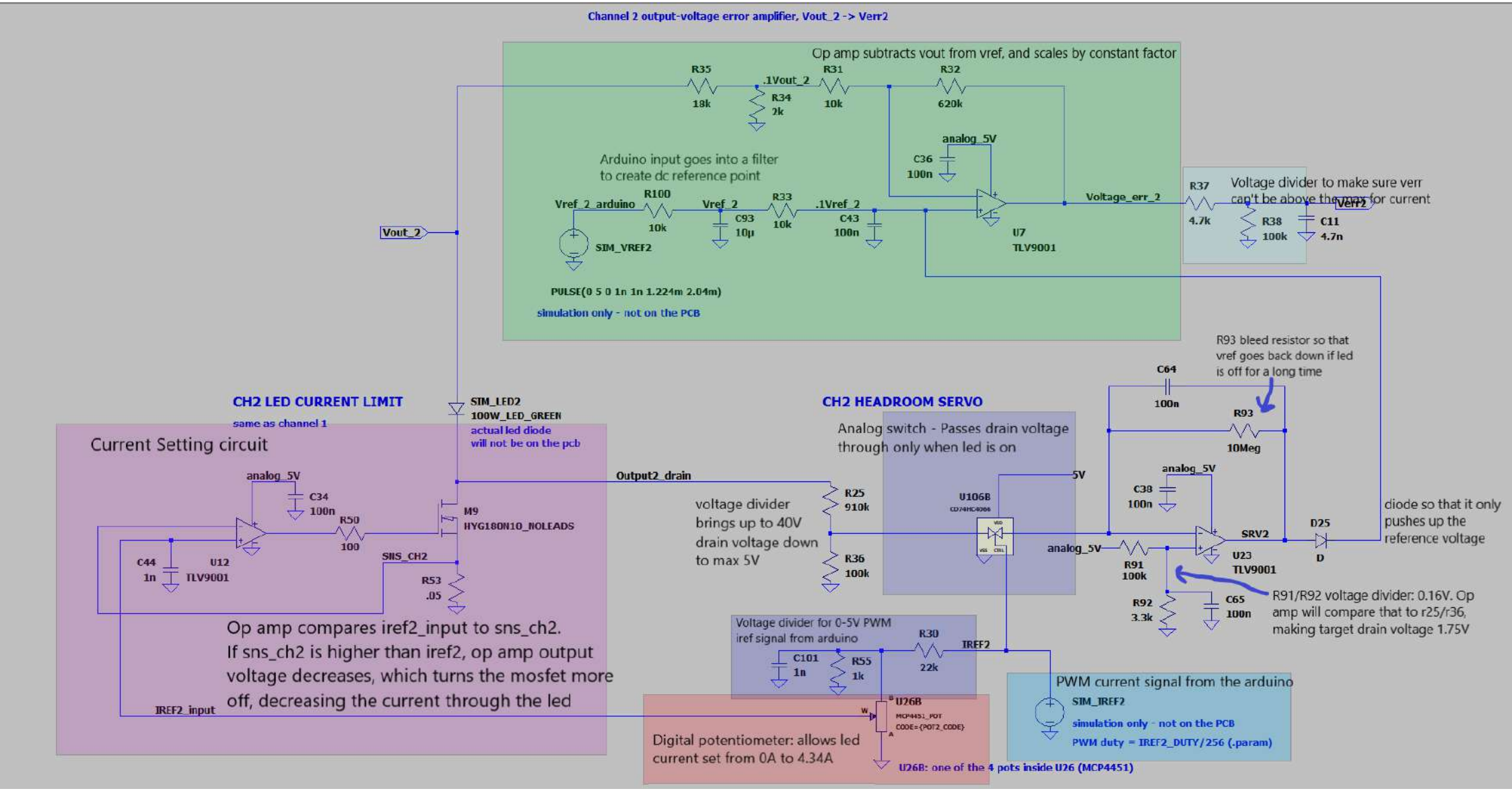
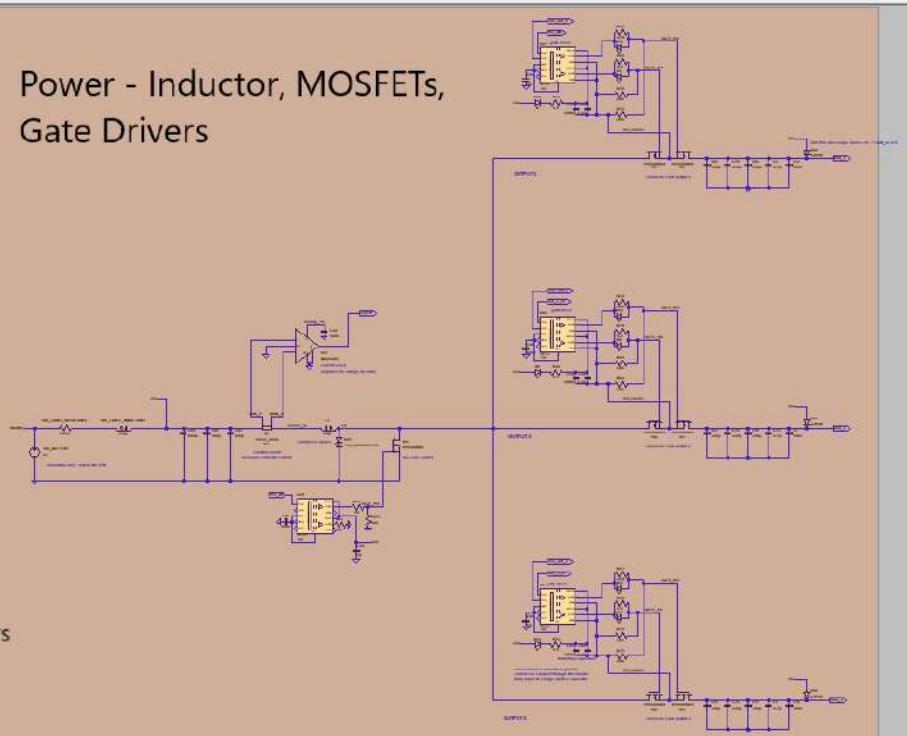

disables circuit if 12V rail is below 9.8V
reenabled at 10.3V
M1 INHIBIT
inhibits m1 if 12V rail is below 9.8V
disables m1 if there is 0 error on the current channel
0err_hi
M1_INHIBIT
M1_ON
U104A
rc circuit delays m1 turn on diode creates fast turn off
Signal that can be sent from the arduino to disable the circuit
ARD_M1_INHIBIT
SIM_ARD_INH
simulation only - not on the PCB
disables circuit if arduino sends disable signal
CHANNEL ENABLE
OUT_1_ENA
OUT_2_ENA
OUT_3_ENA
ena_out_1
ena_out_2
ena_out_3
goes to gate driver
M2_ON
out_2_en
out_3_en
LOW-ERROR TIMEOUT
makes it so that if there is one channel with low error, it doesn't just stop on that one channel.
low_error_timeout
SEQUENCER / ERROR SELECT
CD40178
74HC4051
counter - increments output when out goes high (finishes charging an output)
also increments if low error timeout is fired
MUX - Selects which verr signal to use based on counter
V err
Verr1
Verr2
Verr3
M1 ON/OFF LATCH
0 error signal: output goes high when verr is below threshold
used so that circuit doesn't keep adding more energy to output
used to send signal to switch from discharging to charging
used to send signal to switch from charging to discharging
Current
6A_hi
Reset_raw
SR-Latch
RESET-
out = high -> inductor to ground
out = low -> inductor to output
out- = inverted out

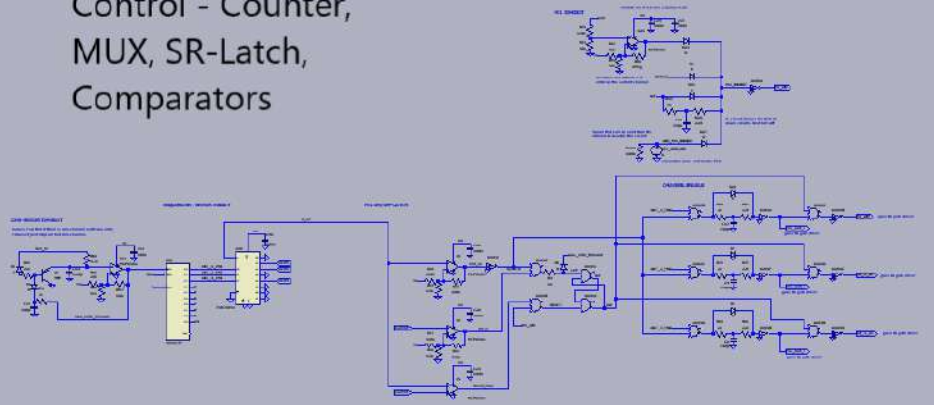

Output 1 Gate Driver
Output 2 Gate Driver
Output 3 Gate Driver
Current sensing
100x (1A = 1V)
battery+cable(simulated)
Input
Capacitors
Clamp diode
OUTPUT 1
OUTPUT 2
OUTPUT 3
connects LX to output 1
connects LX to output 2
connects LX to output 3
low side switch
simulation only - not on the PCB
current sense
amplifies R1 voltage by 100x
bootstrap capacitors
when m1 pulls inductor to ground,
current can conduct through the mosfet
body diode to charge up this capacitor
D28-D30: pre-charge diodes Vin -> Vout_1/2/3


Channel 2 output-voltage error amplifier, Vout_2 -> Verr2
Op amp subtracts vout from vref, and scales by constant factor
R35
18k
.1Vout_2
R34
2k
R31
10k
R32
620k
analog_5V
C36
100n
Arduino input goes into a filter
to create dc reference point
Vref_2_arduino
R100
10k
Vref_2
C93
10µ
R33
10k
.1Vref_2
C43
100n
SIM_VREF2
PULSE(0 5 0 1n 1n 1.224m 2.04m)
simulation only - not on the PCB
U7
TLV9001
Voltage_err_2
R37
4.7k
Voltage divider to make sure verr
can't be above the max for current
R38
100k
C11
4.7n
Verr2
Vout_2
R93 bleed resistor so that
vref goes back down if led
is off for a long time
C64
100n
R93
10Meg
CH2 LED CURRENT LIMIT
same as channel 1
SIM_LED2
100W_LED_GREEN
actual led diode
will not be on the pcb
Current Setting circuit
analog_5V
C34
100n
R50
100
M9
HYG180N10_NOLEADS
C44
1n
U12
TLV9001
SNS_CH2
R53
.05
Output2_drain
CH2 HEADROOM SERVO
Analog switch - Passes drain voltage
through only when led is on
voltage divider
brings up to 40V
drain voltage down
to max 5V
R25
910k
R36
100k
U106B
CD74HC4066
5V
analog_5V
C38
100n
SRV2
D25
D
diode so that it only
pushes up the
reference voltage
analog_5V
R91
100k
U23
TLV9001
R92
3.3k
C65
100n
R91/R92 voltage divider: 0.16V. Op
amp will compare that to r25/r36,
making target drain voltage 1.75V
Op amp compares iref2_input to sns_ch2.
If sns_ch2 is higher than iref2, op amp output
voltage decreases, which turns the mosfet more
off, decreasing the current through the led
Voltage divider for 0-5V PWM
iref signal from arduino
C101
1n
R55
1k
R30
22k
IREF2
IREF2_input
U26B
MCP4451_POT
CODE={POT2_CODE}
Digital potentiometer: allows led
current set from 0A to 4.34A
U26B: one of the 4 pots inside U26 (MCP4451)
PWM current signal from the arduino
SIM_IREF2
simulation only - not on the PCB
PWM duty = IREF2_DUTY/256 (.param)

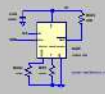
Power - Inductor, MOSFETs, Gate Drivers



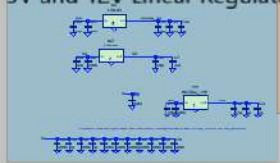
Control - Counter, MUX, SR-Latch, Comparators



Chip to control digital potentiometer (will be connected to arduino)

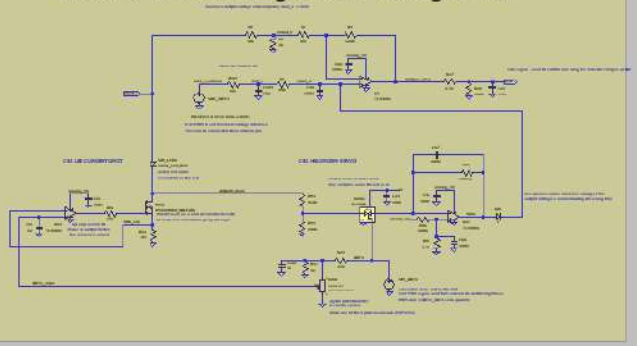


5V and 12V Linear Regulators

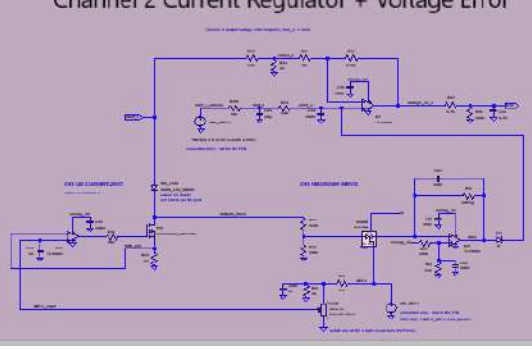




Channel 1 Current Regulator + Voltage Error



Channel 2 Current Regulator + Voltage Error



Channel 3 Current Regulator + Voltage Error

